

材料系 材料力学B 2022春 期中考试

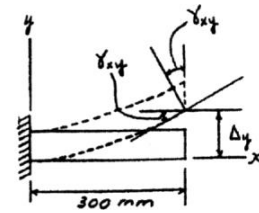
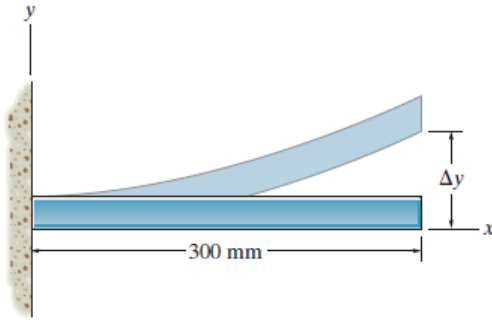
疫情原因考试在家进行，开卷考试。第2题得用Excel做。

Midterm Homework of Mechanics of Materials B

Name:

Student Number:

1. The bar is originally 300 mm long when it is flat. If it is subjected to a shear strain defined by $\gamma_{xy} = 0.02x$, where x is in meters, determine the displacement at the end of its bottom edge. It is distorted into the shape shown, where no elongation of the bar occurs in the x direction. (10 points)



Shear Strain:

$$\frac{dy}{dx} = \tan \gamma_{xy} = \tan (0.02x)$$

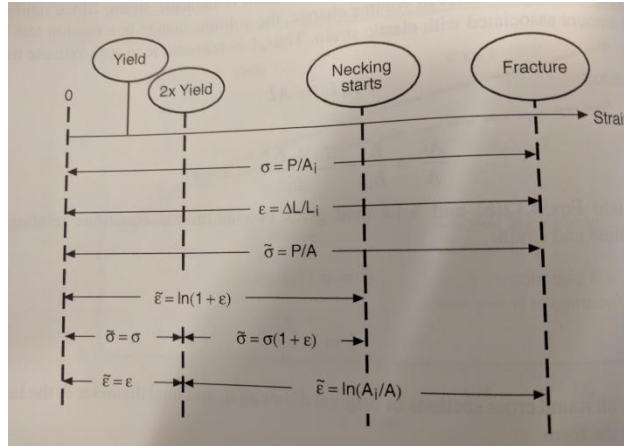
$$\begin{aligned}\Delta y &= \int_0^{0.3m} \tan(0.02x) dx \\ &= -50[\ln \cos(0.02x)] \Big|_0^{0.3m} \\ &= 9 \times 10^{-4} m = 0.9 \text{ mm}\end{aligned}$$

Note: 采用近似原则, $\tan(0.02x) \sim 0.02x$, $\Delta y = 0.9 \text{ mm}$ 。

2. For a number of points during a tension test on Man-Ten steel, engineering stress and strain data are given in Table. Also given are minimum diameters measured in the necked region in the latter portions of the test. The initial diameter was 6.32 mm. (a) Determine the Young's modulus; (b) Determine the true stress and strain throughout; (c) Calculate the true plastic strain for the beyond yielding. (30 points)

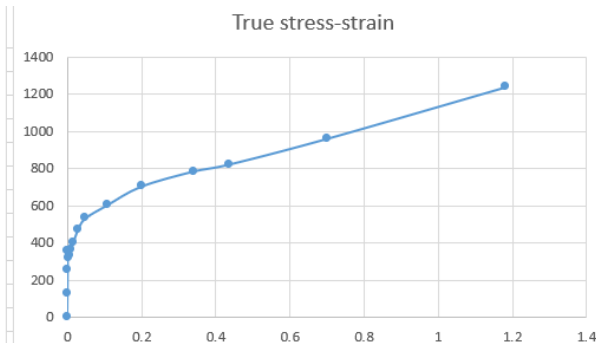
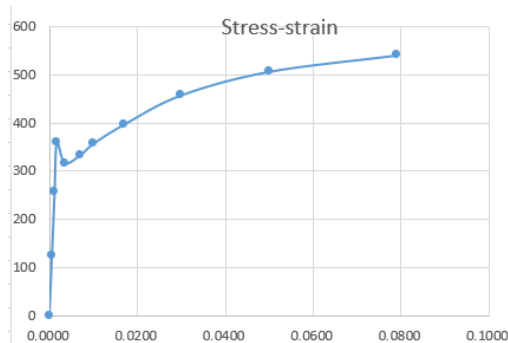
Engr. Stress σ , MPa	Engr. Strain ϵ	Diameter d , mm
0	0	6.32
125	0.0006	—
257	0.0012	—
359	0.0017	—
317	0.0035	—
333	0.0070	—
357	0.0100	—
397	0.0170	—
458	0.0300	—
507	0.0500	—
541	0.0790	5.99
576	—	5.72
558	—	5.33
531	—	5.08
476	—	4.45
379	—	3.50

(Final point is fracture.)

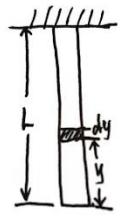
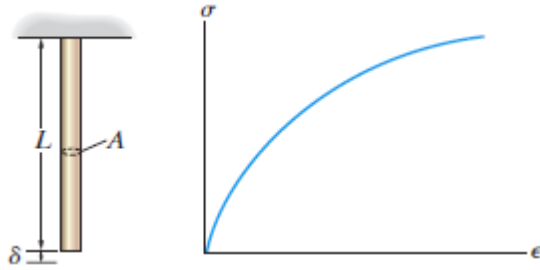


Stress (Mpa)	Strain	Diameter (mm)		True stress (MPa)	True strain	True plastic strain	
0	0.0000	6.32		0	0	0.0000	
125	0.0006			125	0.0006	0.0000	
257	0.0012			257	0.0012	0.0000	
359	0.0017			359	0.0017	0.0000	
317	0.0035			317	0.0035	0.0020	
333	0.0070			333	0.007	0.0054	
357	0.0100			360.57	0.00995	0.0082	
397	0.0170		$\bar{\sigma} = \sigma(1 + \epsilon)$	403.75	0.01686	0.0149	
458	0.0300		$\bar{\epsilon} = \ln(1 + \epsilon)$	471.74	0.02956	0.0273	
507	0.0500			532.35	0.04879	0.0463	
541	0.0790	5.99		602.25	0.10726	0.1044	
576	—	5.72	$\bar{\sigma} = \sigma \frac{d_0^2}{d^2}$	703.18	0.19950	0.1962	
558	—	5.33		784.54	0.34074	0.3370	
531	—	5.08		821.87	0.43682	0.4329	
476	—	4.45	$\bar{\epsilon} = 2 \ln \frac{d_0}{d}$	960.11	0.70163	0.6971	
379	—	3.5		1235.77	1.18191	1.1761	

$$E = \frac{\sigma}{\epsilon} = \frac{357 \text{ MPa}}{0.0017} = 211 \text{ GPa}$$



3. A material has a stress–strain diagram that can be described by the curve $\sigma = c\epsilon^{1/2}$, Determine the deflection δ of the end of a rod made from this material if it has a length L , cross-sectional area A , and a specific weight γ . (20points).



$$\sigma = \frac{P}{A} = \frac{Ay \cdot \gamma}{A} = y \cdot \gamma$$

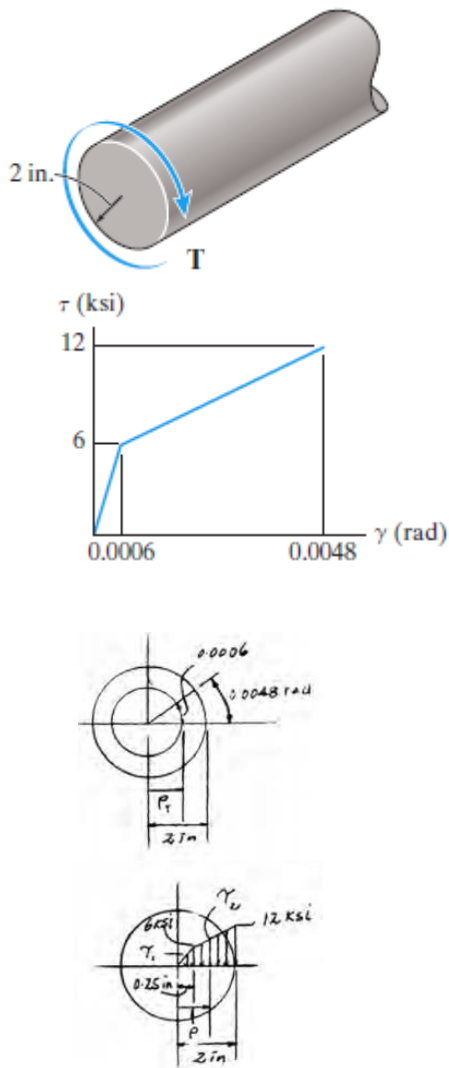
$$\sigma = c\epsilon^{\frac{1}{2}} \leftrightarrow \epsilon = \left(\frac{\sigma}{c}\right)^2$$

$$d\delta = \epsilon dy$$

$$\delta = \int_0^L \epsilon dy = \int_0^L \left(\frac{y \cdot \gamma}{c}\right)^2 dy$$

$$= \frac{\gamma^2 L^3}{3c^2}$$

4. The shaft is subjected to a maximum shear strain of 0.0048 rad. Determine the torque applied to the shaft if the material has strain hardening as shown by the shear stress–strain diagram (20 points).



From the shear - strain diagram,

$$\frac{\rho_Y}{0.0006} = \frac{2}{0.0048}; \quad \rho_Y = 0.25 \text{ in.}$$

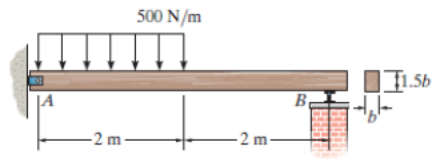
From the shear stress–strain diagram,

$$\tau_1 = \frac{6}{0.25}\rho = 24\rho$$

$$\frac{\tau_2 - 6}{\rho - 0.25} = \frac{12 - 6}{2 - 0.25}; \quad \tau_2 = 3.4286\rho + 5.1429$$

$$\begin{aligned} T &= 2\pi \int_0^c \tau \rho^2 d\rho \\ &= 2\pi \int_0^{0.25} 24\rho^3 d\rho + 2\pi \int_{0.25}^2 (3.4286\rho + 5.1429)\rho^2 d\rho \\ &= 2\pi \left[6\rho^4 \right]_0^{0.25} + 2\pi \left[\frac{3.4286\rho^4}{4} + \frac{5.1429\rho^3}{3} \right]_{0.25}^2 \\ &= 172.30 \text{ kip} \cdot \text{in.} = 14.4 \text{ kip} \cdot \text{ft} \end{aligned}$$

5. The wood beam has a rectangular cross section in the proportion shown. Determine its required dimension b if the allowable bending stress is $\sigma_{\text{allow}} = 10 \text{ MPa}$ (20 points).



$$+\uparrow \sum F_y = 0; \quad F_A + F_B - 500 \times 2 = 0$$

$$\zeta + \sum M_A = 0; \quad -500 \times 2 \times 1 + F_B \times 4 = 0$$

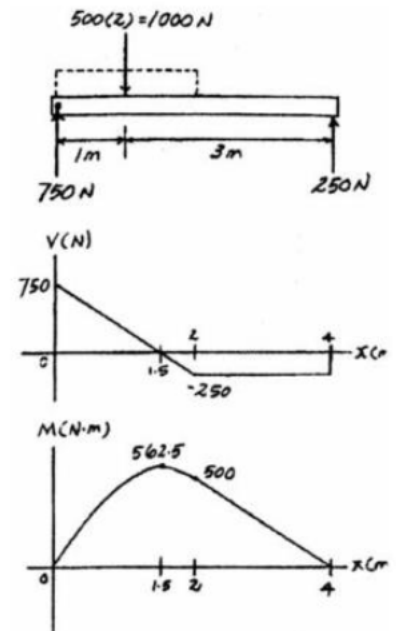
$$F_A = 750 \text{ N}, F_B = 250 \text{ N}$$

Allowable Bending Stress: The maximum moment is $M_{\text{max}} = 562.5 \text{ N} \cdot \text{m}$ as indicated on the moment diagram. Applying the flexure formula

$$\sigma_{\text{max}} = \sigma_{\text{allow}} = \frac{M_{\text{max}} c}{I}$$

$$10(10^6) = \frac{562.5(0.75b)}{\frac{1}{12}(b)(1.5b)^3}$$

$$b = 0.05313 \text{ m} = 53.1 \text{ mm}$$



Ans.